NORTHERN ARIZONA UNIVERSITY

School of Informatics, Computing, and Cyber Systems

CS 212 - Project Phase – 2 Report

Follow the instructions provided in the grey text. Replace the grey text with actual information.

Group Details:

Student 1 Name	Daniel Raygoza
Student 1 Email	dr2646@nau.edu
Student 2 Name	Aaron Skol
Student 2 Email	ars2358@nau.edu
Student 3 Name	Derrick Iron
Student 3 Email	dki4@nau.edu

Phase-2 Details:

Project Title	Course Enrollment Portal
Phase - 2 Components	*My Enrollments Page.* *Override Request Page.* *Course Details Page.* *Notifications Page.* Complete endpoints for: - POST /api/students/new - POST /api/students/:id/waitlist/:id - DELETE /api/students/:id/waitlist/:id - GET /api/students/:id/notifications - GET /api/students/:id/notifications/unread - POST /api/override - GET /api/students/:id/map - [INTERNAL] /api/override/accept - [INTERNAL] /api/override/deny

Task Status:

Avoid incorporating generic information, as it may result in penalties. Copy all the phase-1 tasks listed in the proposed project plan.

Task No	Task Details	Assigned To	Task Status
1	Create the My Enrollments page, fetching the enrolled courses, listing them, and allowing the user to modify them.	Daniel	Completed
2	Create the Override Request page. If the user's prerequisites aren't met for a given course (verified by the backend), then allow the user to send an override to the backend. The form will contain fields like: name, email, course, and reason. With some fields being autopopulated if navigated to this page from a Course Details page.	Daniel	Completed
3	Create the Course Details page. Display information about a course, such as the credits, name, subject, course number, session, career, section, days and times, room, instructor, and whether the course is still open, can be waitlisted, or is closed.	Daniel	Completed
4	Create the Notifications page. Show previous notifications that haven't been read. The main notifications would be if a course was successfully enrolled or dropped, an override was approved	Daniel	Completed

	or denied, or a waitlisted course became enrolled.		
5	Take the parsed Ajax calls and implement the functionality that occurs on success and on error for each one.	Daniel	Completed
6	Implement POST /api/students/:id/waitlist/:id. This will add a student to a waitlist for a specific course ID. This will also trigger a notification to be added to the student's data.	Aaron	Completed
7	Implement DELETE /api/students/:id/waitlist/:id. This will delete a student from a waitlist for a specific course ID. This will also trigger a notification to be added to the student's data.	Aaron	Completed
8	Implement a system that locks the database whenever it’s about to be modified to prevent multiple writes at the same time. Then, when it unlocks, do the next request and repeat.	Aaron	Completed
9	Implement GET /api/students/:id/map. Fetching the students’ current enrolled courses map based on their student ID to be displayed on the Course Map Component. Allowing for a body to the request to contain {additional_course_ids: []} to display.	Aaron	Completed
10	Write the frontend parser that checks the pathname and extracts the query params, which then calls the correct JQuery Ajax calls.	Aaron	Completed/ Backlog (Pending)/ Modified
11	Implement GET /api/students/:id/notifications. This will fetch all notifications for a specific student.	Derrick	Completed

	Implement GET /api/students/:id/notifications/unread/ This will fetch how many unread notifications the student has.		
12	Implement POST /api/override. Taking in the body: {id: 0, name: “”, email: “”, sid: ”“, course_id: “”, reason: “”, date: “”}. Store this request in an override request JSON database to be evaluated.	Derrick	Completed
13	Implement a private endpoint POST /api/override/accept Which accepts an override request ID and accepts it. Updating the courses database and the students database. Then, send out a notification to the user informing them.	Derrick	Completed
14	Implement a private endpoint POST /api/override/deny Which accepts an override request ID and denies it. Updating the courses database and the students database. Then, send out a notification to the user informing them.	Derrick	Completed
15	For each subject in the JSON database, assign them a location, along with creating a coordinates map for each location.	Derrick	Completed

New Tasks or Modified Tasks (Optional):

List all newly created tasks or tasks modified from your project proposal. Feel free to add more rows to the following table. **Fill the below section, only if you have any new tasks or modifications in the Phase-2 tasks list or else skip it.**

Task Type	Task Details	Assigned To	Task Status

New/ Modified	Implement the API endpoint: POST /api/students/new; allowing for new users to be created if the user is not already present in the database.	Daniel	Completed
New/ Modified	Explain the Task in detail (not in general)	Name of the team member	Completed/ Backlog (Pending)/ Modified

Challenges and Solutions Implementation:

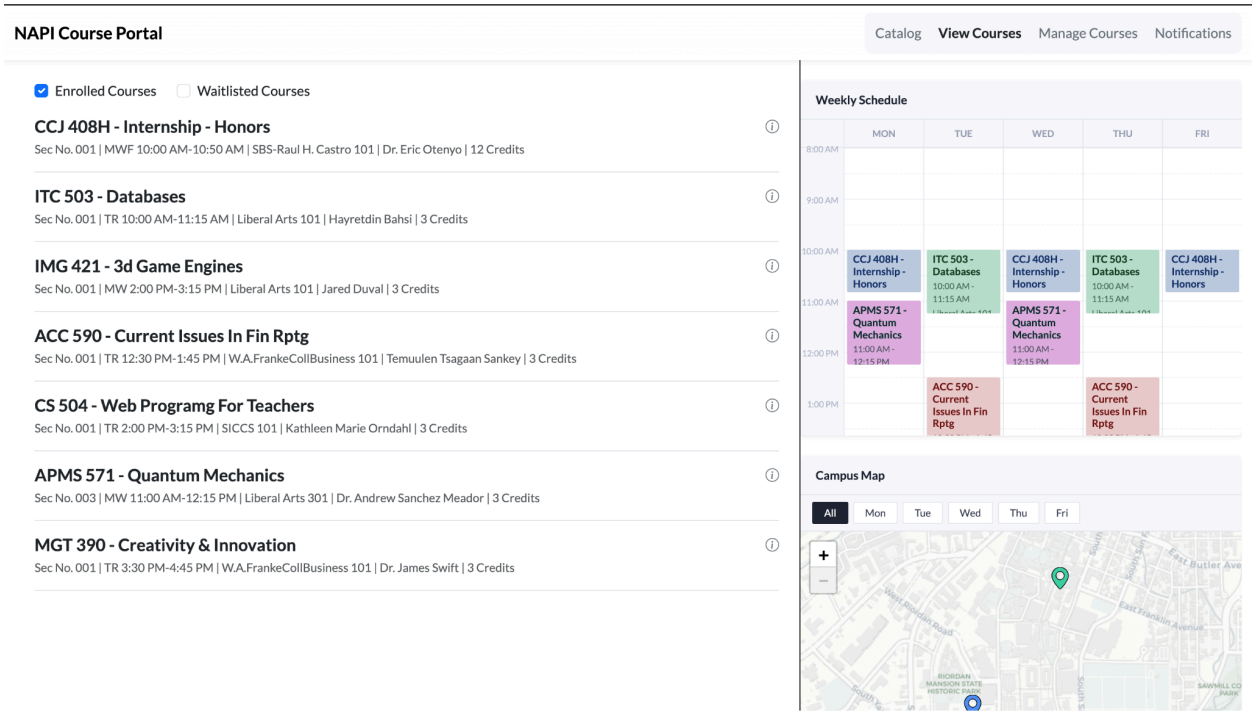
List at least 4 challenges faced by your team during phase-2 development and elaborate on the solutions you have implemented. Feel free to add more rows to the following table.

Details of the Challenge	Solution Implemented
During Phase 2, I had a struggle with where my code fit in for the notifications and had to copy and paste into courses.js instead of studentmodel.js	So I had to open all the JS files and use trial and error, and eventually found where they belonged.
When making the overrides page, I realized that just using text inputs to select the course would allow the user to accidentally select bad inputs.	Reflecting on phase 1, developing the course selector for the catalog page, I realized I can bring in much of that code and modify it slightly to get an improved course selector for users.
The notifications lacked some extra data that was needed for the frontend.	We had to go through each service to properly implement IDs and notification types so that the frontend could display the proper data.
When implementing the route for the map, the path was inconsistent with what was defined within /server.js. This among other inconsistencies with definitions and naming conventions made debugging tedious.	By testing the endpoints in the front-end, it becomes clear where data fails to be initialized and we can then locate the source file with the undefined instance.

Phase-2 Developed Screens:

- Incorporate webpage screens corresponding to the project phase - 2.
- Capture all screenshots of your phase-2 delivery (**don't include any code screenshots**) and attach them in this section.

[**Attach all the screenshots of the phase-2 development webpage screens here**]



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NAPI Course Portal

Catalog | View Courses | Manage Courses | Notifications

CCJ 408H - Internship - Honors [001]

Class Details

Subject	Course Number	Credits	Level
CCJ	408H	12	UNDERGRADUATE

Meeting Info

Section	Days & Times	Room	Instructor
001	MWF 10:00 AM - 10:50 AM	SBS-Raul H. Castro 101	Dr. Eric Otenyo

Enrollment Info

Prerequisite(s):

- CCJ-Criminology & Criminal Justice GPA of 3.0
- better
- Honors Student Group

Class Availability

Open 20 / 33 enrolled

Enroll | Request Override

Weekly Schedule

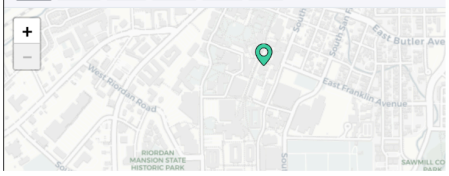
Schedule Conflict(s) Found

Conflict between Courses:
CCJ 408H - Internship - Honors (MWF 10:00 AM-10:50 AM) and
CCJ 408H - Internship - Honors (MWF 10:00 AM-10:50 AM)

	MON	TUE	WED	THU	FRI
8:00 AM					
9:00 AM					
10:00 AM	CCJ 408H - Internship - Honors	ITC 503 - Databases 10:00 AM -	CCJ 408H - Internship - Honors	ITC 503 - Databases 10:00 AM -	CCJ 408H - Internship - Honors

Campus Map

All | Mon | Tue | Wed | Thu | Fri



NAPI Course Portal

Catalog | View Courses | **Manage Courses** | Notifications

Enrolled Courses

- CCJ 408H - Internship - Honors — DROP
- ITC 503 - Databases — DROP
- IMG 421 - 3d Game Engines — DROP
- ACC 590 - Current Issues In Fin Rptg — DROP
- CS 504 - Web Programg For Teachers — DROP
- APMS 571 - Quantum Mechanics — DROP
- MGT 390 - Creativity & Innovation — DROP

Waitlisted Courses

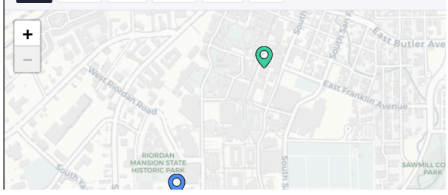
- PM 686 - Advanced Project Management — Remove Waitlist

Weekly Schedule

	MON	TUE	WED	THU	FRI
8:00 AM					
9:00 AM					
10:00 AM	CCJ 408H - Internship - Honors	ITC 503 - Databases 10:00 AM - 11:15 AM	CCJ 408H - Internship - Honors	ITC 503 - Databases 10:00 AM - 11:15 AM	CCJ 408H - Internship - Honors
11:00 AM	APMS 571 - Quantum Mechanics 11:00 AM - 12:15 PM		APMS 571 - Quantum Mechanics 11:00 AM - 12:15 PM		
12:00 PM					
1:00 PM		ACC 590 - Current Issues In Fin Rptg		ACC 590 - Current Issues In Fin Rptg	

Campus Map

All | Mon | Tue | Wed | Thu | Fri



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NAPI Course Portal

Catalog | View Courses | Manage Courses | **Notifications**

Successful Enrollement — 4/27/2026, 1:02:40 PM
You are now enrolled in the course: UNDERSTANDING POLITICAL INQUIRY

Course Successfully Dropped — 4/27/2026, 12:56:11 PM
You are no longer in the course: INTERNSHIP - HONORS

Course Successfully Dropped — 4/27/2026, 12:56:10 PM
You are no longer in the course: DATABASES

Course Successfully Dropped — 4/27/2026, 12:56:05 PM
You are no longer in the course: 3D GAME ENGINES